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United States Patent	12397473
Kind Code	B2
Date of Patent	August 26, 2025
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Boring device

Abstract

A boring device according to one embodiment of the present invention includes: a cutting edge; a drive unit that rotationally drives the cutting edge; a feed flow path through which a cooling liquid is fed to the cutting edge; and a tube pump that forces the cooling liquid in the feed flow path to flow to the cutting edge. The tube pump includes a feed tube constituting a part of the feed flow path and a first pressing mechanism that presses the feed tube, and the first pressing mechanism is driven by the drive unit.

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Family ID:	1000008781840
Appl. No.:	17/905477
Filed (or PCT Filed):	February 22, 2021
PCT No.:	PCT/JP2021/006611
PCT Pub. No.:	WO2021/199791
PCT Pub. Date:	October 07, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230118075 A1	Apr. 20, 2023

Foreign Application Priority Data

JP	2020-062957	Mar. 31, 2020
JP	2020-151051	Sep. 09, 2020

Publication Classification

Int. Cl.: B28D7/02 (20060101); B28D1/14 (20060101); F04B43/12 (20060101)

U.S. Cl.:

CPC B28D7/02 (20130101); B28D1/14 (20130101); F04B43/1253 (20130101);

Field of Classification Search

CPC: B28D (7/02); B28D (1/14); F04B (43/1253); F16H (1/16)

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to boring devices.

BACKGROUND ART

(2) For example, a boring device as proposed in Patent Literature 1 is conventionally known.

(3) The boring device of Patent Literature 1 includes a drill bit and an electric drill that rotates the drill bit. The boring device of Patent Literature 1 further includes: a cooling liquid feed attachment disposed between the drill bit and the electric drill to feed a cooling liquid to a cutting edge of the drill bit; and a cooling liquid tank that holds the cooling liquid. The cooling liquid tank is provided with a pump, by which the cooling liquid held in the cooling liquid tank is forced to flow to the cooling liquid feed attachment through a feed flow path.

CITATION LIST

SUMMARY OF INVENTION

Technical Problem

(5) The configuration of the boring device of Patent Literature 1 is complicated because the cooling liquid tank is provided with the pump that forces the cooling liquid to flow to the cutting edge of the drill bit.

(6) It is therefore an object of the present invention to provide a boring device capable of forcing a cooling liquid to flow to a cutting edge with a simple device configuration.

Solution to Problem

(7) In order to solve the problem described above, a boring device according to the present invention includes: a cutting edge; a drive unit that rotationally drives the cutting edge; a feed flow path through which a cooling liquid is fed to the cutting edge; and a tube pump that forces the cooling liquid in the feed flow path to flow to the cutting edge, wherein the tube pump includes a feed tube constituting a part of the feed flow path and a first pressing mechanism that presses the feed tube, and the first pressing mechanism is driven by the drive unit.

(8) With the above features, the tube pump is driven by the drive unit that rotationally drives the cutting edge. Thus, the cooling liquid can be forced to flow to the cutting edge with a simple device configuration.

Advantageous Effects of Invention

(9) The present invention can provide a boring device capable of forcing a cooling liquid to flow to a cutting edge with a simple device configuration.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view showing the overall configuration of a boring device according to a first embodiment of the present invention.

(2) FIG. 2 is a perspective view of a pump unit of the boring device according to the first embodiment of the present invention and shows the pump unit as viewed from a drive unit.

(3) FIG. 3 is a perspective view of a planetary gear mechanism closest to the drive unit among a plurality of planetary gear mechanisms of the boring device according to the first embodiment of the present invention and shows the planetary gear mechanism as viewed in the course of assembly.

(4) FIG. 4 is a perspective view of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms of the boring device according to the first embodiment of the present invention and shows the planetary gear mechanism with a planetary carrier attached thereto.

(5) FIG. 5 is a perspective view of a planetary gear mechanism closest to a cutting edge among the plurality of planetary gear mechanisms of the boring device according to the first embodiment of the present invention and shows the planetary gear mechanism with pressers attached to a plurality of planetary gears of the planetary gear mechanism.

(6) FIG. 6 is a perspective view of the planetary gear mechanism closest to the cutting edge among the plurality of planetary gear mechanisms of the boring device according to the first embodiment of the present invention and shows the planetary gear mechanism with a third circular plate and feed tube attached thereto.

(7) FIG. 7 is a perspective view of the pump unit of the boring device according to the first embodiment of the present invention and shows the pump unit as viewed from the cutting edge.

(8) FIG. 8 is a schematic view showing the overall configuration of a boring device according to a second embodiment of the present invention.

- (9) FIG. 9 is a schematic view showing the overall configuration of a boring device according to a third embodiment of the present invention.
- (10) FIG. 10 is a schematic view showing the overall configuration of a boring device according to a fourth embodiment of the present invention.
- (11) FIG. 11 is a cross-sectional view of a cooling liquid feed mechanism and its neighboring components of the boring device according to the fourth embodiment of the present invention and shows the cooling liquid feed mechanism and its neighboring components cut at the center in the width direction of the boring device along the longitudinal direction of the boring device.
- (12) FIGS. 12A and 12B are each a schematic view of a shield of the boring device according to the fourth embodiment of the present invention; FIG. 12A is a schematic view of the shield as viewed from the proximal end of a drill bit, and FIG. 12B is a cross-sectional view of the shield cut at the center in the width direction of the boring device along the longitudinal direction of the boring device.
- (13) FIG. 13 is a schematic view showing a biasing mechanism of the boring device according to the fourth embodiment of the present invention.
- (14) FIGS. 14A and 14B are each a cross-sectional view of the cooling liquid feed mechanism and its neighboring components of the boring device according to the fourth embodiment of the present invention and show the cooling liquid feed mechanism and its neighboring components cut at the center in the width direction of the boring device along the longitudinal direction of the boring device; FIG. 14A is a cross-sectional view of the cooling liquid feed mechanism and its neighboring components in a state with the feed flow path closed before a boring operation, and FIG. 14B is a cross-sectional view of the cooling liquid feed mechanism and its neighboring components in a state with the feed flow path open during the boring operation.

DESCRIPTION OF EMBODIMENTS

(15) Hereinafter, boring devices according to exemplary embodiments of the present invention will be described with reference to the drawings. The present invention is not limited by the described embodiments. In the following description, the same or like elements are denoted by the same reference signs throughout the drawings and will not be described repeatedly.

First Embodiment

- (16) FIG. 1 is a schematic view showing the overall configuration of a boring device according to a first embodiment. As shown in FIG. 1, the boring device 10A according to the present embodiment includes an electric drill 11, a pump unit 20 attached to the electric drill 11, and a drill bit 90 attached to the electric drill 11 via the pump unit 20. The boring device 10A further includes a feed flow path 88 through which a cooling liquid L is fed to a cutting edge 98 of the drill bit 90.
- (17) In the present embodiment, the drill bit 90 is rotationally driven by the electric drill 11, and the cutting edge 98 of the rotating drill bit 90 is pressed against a workpiece such as a concrete or stone material (not shown) to perform a boring operation. During such a boring operation, the cooling liquid L in the feed flow path 88 is forced by the pump unit 20 to flow to the cutting edge 98 of the drill bit 90 for purposes such as reducing the heat generation of the drill bit 90 and the workpiece.
- (18) The drill bit 90 is hollow. Thus, the cooling liquid L flows through the interior of the drill bit 90 and is fed to the cutting edge 98 of the drill bit 90. The cooling liquid L fed to the cutting edge 98 of the drill bit 90 is discharged out of the cutting edge 98 to reduce the heat generation of the drill bit 90 and the workpiece. The cooling liquid L mixes with chips of the workpiece to form a suspension, which is discharged outside.
- (19) A cooling liquid tank (not shown) that holds the cooling liquid L may be disposed at the proximal end of the feed flow path 88. The cooling liquid L is typically water and may be, for example, another liquid having a low viscosity. For example, when the workpiece is an iron plate, an oil having a low viscosity may be used as the cooling liquid L.
- (20) Electric Drill 11
- (21) The electric drill 11 includes a housing 12, a drive unit 13 enclosed in the housing 12 to

rotationally drive the drill bit **90**, and a first gripping mechanism **18** disposed outside the housing **12** to grip a shank **22** projecting from the proximal end surface of the pump unit **20**.

(22) The drive unit **13** includes an electric motor **14** and a motor speed reduction mechanism **16** attached to a drive shaft **15** of the electric motor **14** to reduce the speed of rotation produced by power derived from the electric motor **14** and transmit the resulting power to the shank **22**.

(23) Pump Unit **20**

(24) FIG. **2** is a perspective view of the pump unit of the boring device according to the present embodiment and shows the pump unit as viewed from the drive unit. As shown in FIG. **2**, the pump unit **20** includes a tube pump **60** that forces the cooling liquid **L** in the feed flow path **88** to flow to the cutting edge **98** of the drill bit **90** and a speed reduction mechanism **30** disposed between the drive unit **13** and the tube pump **60** (or a first pressing mechanism **64** described later) to reduce the speed of rotation produced by power derived from the drive unit **13** and transmit the resulting power to the tube pump **60** (or the first pressing mechanism **64**).

(25) The pump unit **20** further includes an input shaft **21** through which the power derived from the drive unit **13** is input to the speed reduction mechanism **16**. The input shaft **21** extends from the drive unit **13** and penetrates the pump unit **20** toward the cutting edge **98**. The pump unit **20** further includes a second gripping mechanism **80** disposed at the distal end of the input shaft **21** to grip a proximal end portion of the drill bit **90**. The input shaft **21** is connected to the drill bit **90** via the second gripping mechanism **80** and rotates together with the drill bit **90**. The proximal end portion of the input shaft **21** is the shank **22** described above.

(26) Speed Reduction Mechanism **30**

(27) The speed reduction mechanism **30** includes two planetary gear mechanisms **31a** and **31b** aligned in the axial direction of the input shaft **21**. The planetary gear mechanism **31a** is located toward the drive unit **13** (this planetary gear mechanism **31a** corresponds to “planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms” or “one of the adjacent planetary gear mechanisms that is closer to the drive unit”). The planetary gear mechanism **31b** is located toward the drill bit **90** (this planetary gear mechanism **31b** corresponds to “planetary gear mechanism closest to the cutting edge among the plurality of planetary gear mechanisms” or “the other of the adjacent planetary gear mechanisms that is closer to the cutting edge”).

(28) FIG. **3** is a perspective view of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms of the boring device according to the present embodiment and shows the planetary gear mechanism as viewed in the course of assembly. As shown in FIG. **3**, the speed reduction mechanism **30** further includes a first circular plate **32**. At the center of the first circular plate **32** is located a through hole **33** (see FIG. **2**) through which the input shaft **21** extends. The edge surface of the first circular plate **32** and the major surface of the first circular plate **32** that faces toward the drive unit **13** define a part of the outer shape of the pump unit **20**.

(29) The planetary gear mechanism **31a** includes a sun gear **34a** located at the center of the first circular plate **32**, three planetary gears **36a** externally contacting the sun gear **34** to rotate in mesh with the sun gear **34a**, and an internal gear **38a** internally contacted by the three planetary gears **36a** to rotate in mesh with the three planetary gears **36a**. The diameter of the internal gear **38a** is equal to the diameter of the first circular plate **32**. The edge surface of the internal gear **38a** defines a part of the outer shape of the pump unit **20**.

(30) The sun gear **34a** projects from the outer surface of the input shaft **21** and is integrated with the outer surface of the input shaft **21**. In other words, the sun gear **34a** is secured to the outer surface of the input shaft **21**. As such, the sun gear **34a** rotates together with the input shaft **21** and is coaxial with the input shaft **21**.

(31) FIG. **4** is a perspective view of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms of the boring device according to the present

embodiment and shows the planetary gear mechanism with a planetary carrier attached thereto. As shown in FIG. 4, the speed reduction mechanism 30 further includes a planetary carrier 50 shaped as a circular plate and connecting the planetary gear mechanisms 31a and 31b together.

(32) The planetary carrier 50 includes three fitting holes 52 (fitting-receiving portions) into which respective rotational shafts 37a of the three planetary gears 36a are fitted. Thus, the planetary carrier 50 is connected to the three planetary gears 36a. The three fitting holes 52 are located near the periphery of the planetary carrier 50 and arranged at regular intervals in the circumferential direction of the planetary carrier 50. The planetary carrier 50 further includes a shaft hole 54 (second shaft hole) through which the input shaft 21 extends. The shaft hole 54 is located at the center of the planetary carrier 50.

(33) A sun gear 34b of the planetary gear mechanism 31b is integrated with the central portion of the major surface of the planetary carrier 50 that faces toward the cutting edge 98. Thus, the planetary carrier 50 rotates together with the sun gear 34b.

(34) FIG. 5 is a perspective view of the planetary gear mechanism closest to the cutting edge among the plurality of planetary gear mechanisms of the boring device according to the present embodiment and shows the planetary gear mechanism with pressers attached to the plurality of planetary gears of the planetary gear mechanism. As shown in FIG. 5, the speed reduction mechanism 30 further includes a second circular plate 56 located closer to the cutting edge 98 than the planetary carrier 50. At the center of the second circular plate 56 is located a through hole 58 through which the input shaft 21 extends and through which the sun gear 34b passes. The diameter of the second circular plate 56 is equal to the diameter of the first circular plate 32 and internal gear 38a. The edge surface of the second circular plate 56 defines a part of the outer shape of the pump unit 20.

(35) The planetary gear mechanism 31a includes the sun gear 34b integrated with the planetary carrier 50 as described above and further includes three planetary gears 36b externally contacting the sun gear 34b to rotate in mesh with the sun gear 34b and an internal gear 38b internally contacted by the three planetary gears 36b to rotate in mesh with the three planetary gears 36b. The sun gear 34b includes a shaft hole 35 (first shaft hole) through which the input shaft 21 extends. The shaft hole 35 communicates with the shaft hole 54 of the planetary carrier 50. The diameter of the internal gear 38b is equal to the diameter of the first circular plate 32, second circular plate 56, and internal gear 38a. The edge surface of the internal gear 38b defines a part of the outer shape of the pump unit 20.

(36) Tube Pump 60

(37) FIG. 6 is a perspective view of the planetary gear mechanism closest to the cutting edge among the plurality of planetary gear mechanisms of the boring device according to the present embodiment and shows the planetary gear mechanism with a third circular plate and feed tube attached thereto. FIG. 7 is a perspective view of the pump unit of the boring device and shows the pump unit as viewed from the cutting edge.

(38) As shown in FIGS. 5 to 7, the tube pump 60 includes a feed tube 62 constituting a part of the feed flow path 88 and a first pressing mechanism 64 that presses the feed tube 62.

(39) As shown in FIGS. 5 and 6, the first pressing mechanism 64 includes three rollers 66 (pressers) mounted on the three planetary gears 36b, respectively, and rotatable together with the three planetary gears 36b, respectively. The three rollers 66 are coaxial with the three planetary gears 36b, respectively, and mounted on the respective rotational shafts 37b of the three planetary gears 36b.

(40) As shown in FIG. 6, the first pressing mechanism 64 further includes a third circular plate 67 located closer to the cutting edge 98 than the three rollers 66. The third circular plate 67 includes three fitting holes 68 into which the respective rotational shafts 37b of the planetary gears 36b are fitted. Thus, the third circular plate 67 is connected to the three planetary gears 36b via the three rollers 66. The three fitting holes 68 are located near the periphery of the third circular plate 67 and

arranged at regular intervals in the circumferential direction of the third circular plate 67. The third circular plate 67 further includes a through hole 69 through which the input shaft 21 extends. The through hole 69 is located at the center of the third circular plate 67.

(41) As shown in FIG. 7, the first pressing mechanism 64 further includes a cover 70 including an inner wall 72 internally contacted by the three rollers 66, with the feed tube 62 interposed between the inner wall 72 and the three rollers 66, to press the feed tube 62 in cooperation with the three rollers 66. The cover 70 is shaped as a hollow circular cylinder and coaxial with the input shaft 21. At the center of one base of the cover 70 that faces toward the cutting edge 98 is located a through hole 73 through which the input shaft 21 extends. The other base of the cover 70 which faces toward the drive unit 13 is open. The inner wall 72 is the inner wall of the circumferential surface of the cover 70. The circumferential surface of the cover 70 is provided with an insertion hole 75 through which the feed tube 62 extending from the cooling liquid tank is inserted into the cover 70 and an outlet hole 76 through which the feed tube 62 is drawn out of the cover 70 toward the cutting edge 98.

(42) The diameter of the cover 70 is equal to the diameter of the first circular plate 32, second circular plate 56, and internal gears 38a and 38b. The base of the cover 70 that faces toward the cutting edge 98 and the circumferential surface of the cover 70 define a part of the outer shape of the pump unit 20.

Effects

(43) In the boring device 10A according to the present embodiment, which is constructed as described above, the tube pump 60 is driven by the drive unit 13 that rotationally drives the cutting edge 98. Thus, the cooling liquid can be forced to flow to the cutting edge 98 with a simple device configuration.

(44) The boring device 10A according to the present embodiment includes the speed reduction mechanism 30, by virtue of which the first pressing mechanism 64 can be rotated at a lower speed than the drill bit 90. This makes it possible, for example, to force the cooling liquid L in the feed flow path 88 to properly flow to the cutting edge 98 of the drill bit 90 while rotating the drill bit 90 at a sufficiently high speed, thus accomplishing a boring operation successfully.

(45) In the present embodiment, the sun gears 34a and 34b are coaxial with the input shaft 21, and this allows for further simplification of the configuration of the boring device 10A.

(46) In the present embodiment, the tube pump 60 is disposed outside the housing 12, and this makes it easier to attach and detach the tube pump 60 to and from the drive unit 13 (or the electric drill 11).

Second Embodiment

(47) A boring device according to a second embodiment of the present invention will be described based on FIG. 8. FIG. 8 is a schematic view showing the overall configuration of the boring device according to the present embodiment. The boring device 10B according to the present embodiment has essentially the same construction as the boring device 10A according to the first embodiment described above, except for the location of the tube pump 60. The same elements will be denoted by the same reference signs, and the same descriptions will not be repeated.

(48) In the present embodiment, as shown in FIG. 8, the tube pump 60 is disposed inside the housing 12. Specifically, the tube pump 60 is located between the drive unit 13 disposed inside the housing 12 and the first gripping mechanism 18 disposed outside the housing 12.

(49) In the present embodiment, the proximal end portion (i.e., the shank 22) of the input shaft 21 of the tube pump 60 is coupled to the motor speed reduction mechanism 16 inside the housing 12. Thus, the motor speed reduction mechanism 16 reduces the speed of rotation produced by power derived from the electric motor 14 and transmits the resulting power to the input shaft 21 of the pump unit 20.

(50) In the present embodiment, a shank 82 projects from the proximal end of the second gripping mechanism 80. The shank 82 is gripped by the first gripping mechanism 18 and connected to the

distal end portion of the input shaft **21** inside the first gripping mechanism **18**.

(51) In the present embodiment constructed as described above, the power derived from the electric motor **14** is transmitted to the drill bit **90**. In the present embodiment, the occurrence of failure of the tube pump **60** can be reduced since the tube pump **60** is enclosed in the housing **12** and not exposed to the outside environment.

Third Embodiment

(52) A boring device according to a third embodiment of the present invention will be described based on FIG. **9**. FIG. **9** is a schematic view showing the overall configuration of the boring device according to the present embodiment. The boring device **10C** according to the present embodiment has the same construction as the boring device **10B** according to the second embodiment, except that the boring device **10C** is devoid of the motor speed reduction mechanism **16**. The same elements will be denoted by the same reference signs, and the same descriptions will not be repeated.

(53) As shown in FIG. **9**, the boring device **10C** according to the present embodiment is devoid of the motor speed reduction mechanism **16**, and the drive shaft **15** of the electric motor **14** serves as the input shaft of the pump unit **20**. This allows, for example, for weight reduction of the boring device **10C**. In the present embodiment, the input shaft of the pump unit **20** may be provided separately from the drive shaft **15** of the electric motor **14**, and the proximal end portion of the input shaft may be attached directly to the distal end portion of the drive shaft **15**.

Fourth Embodiment

(54) A boring device according to a fourth embodiment of the present invention will be described based on FIGS. **10** to **14**. FIG. **10** is a schematic view showing the overall configuration of the boring device according to the present embodiment. The construction of the boring device **10D** according to the present embodiment is identical in many ways to that of the boring device **10B** according to the second embodiment described above. The same elements will be denoted by the same reference signs, and the same descriptions will not be repeated.

(55) As shown in FIG. **10**, the boring device **10D** according to the present embodiment further includes a discharge flow path **138** through which a suspension **S** is discharged from the cutting edge **98**. In the present embodiment, the tube pump **60'** further includes a discharge tube **112** constituting a part of the discharge flow path **138** and a second pressing mechanism **114** that presses the discharge tube **112**.

(56) The circumferential surface of the cover **70** is provided with an insertion hole **75'** through which the discharge tube **112** extending from the cutting edge **98** is inserted into the cover **70**, in addition to the insertion hole **75** for the feed tube **62**. The circumferential surface of the cover **70** is provided with an outlet hole (not shown) through which the discharge tube **112** is drawn out of the cover **70** toward a tank **150** described later, in addition to the outlet hole **76** (not shown in FIG. **10**; see FIG. **7**) for the feed tube **62**. The insertion hole **75'** and outlet hole for the discharge tube **112** are located closer to the cutting edge **98** than the insertion hole **75** and outlet hole **76** for the feed tube **62**, respectively, and are located at the same height as the insertion hole **75** and outlet hole **76**, respectively.

(57) In the present embodiment, a single pressing mechanism **64** (**114**) serves as both the first pressing mechanism **64** that presses the feed tube **62** and the second pressing mechanism **114** that presses the discharge tube **112**. Specifically, the pressing mechanism **64** (**114**) of the present embodiment has the same construction as the first pressing mechanism **64** described above based on FIGS. **5** to **7**. The three rollers **66** (see FIGS. **5** and **6**) and the inner wall **72** (see FIG. **7**) of the cover **70** press the discharge tube **112** as well as the feed tube **62**. Thus, the pressing mechanism **64** (**114**) forces the suspension **S** in the discharge flow path **138** to flow downstream of the discharge flow path **138** in addition to forcing the cooling liquid **L** in the feed flow path **88** to flow to the cutting edge **98**.

(58) As shown in FIG. **10**, the upstream end of the feed flow path **88** and the downstream end of

the discharge flow path **139** are connected to a separation device **140** that separates chips **C** of the workpiece from the suspension **S** to recover the cooling liquid **L**.

(59) Separation Device 140

(60) The separation device **140** includes a liquid vessel **141** and an inner vessel **142** disposed inside the liquid vessel **141**. The separation device **140** further includes a float **146** floating on the suspension **S** held in the inner vessel **142** and a strainer **147** that is movable in the suspension **S** together with the float **146** and that separates the chips **C** from the suspension **S**.

(61) The suspension **S** is discharged from the downstream end of the discharge flow path **138** into the inner vessel **142**, and then the chips **C** and other substances settle under their own weight. Additionally, the strainer **147** removes the chips **C** (chips **C** that are too large to pass between a sliding member **167** and a rotor **161** which are described later or between an incline **167a** and a sealing member **165** which are also described later) in the vicinity of the liquid surface of the suspension **S**. Thus, the cooling liquid **L** recovered by separation of the chips **C** from the suspension **S** can be fed to the upstream end of the feed flow path **88** which is connected to the strainer **147**.

(62) As shown in FIG. **10**, the boring device **10D** according to the present embodiment further includes: a cooling liquid feed mechanism **160** located closer to the cutting edge **98** than the housing **12** and the first gripping mechanism **18**; and a shield **200** that covers a boring target area **Wa** of a workpiece **W** in which a hole is bored by the cutting edge **98** of the drill bit **90**. The shield **200** is biased forward by a biasing mechanism **170** described later.

(63) FIG. **11** is a cross-sectional view of the cooling liquid feed mechanism and its neighboring components of the boring device according to the present embodiment and shows the cooling liquid feed mechanism and its neighboring components cut at the center in the width direction of the boring device along the longitudinal direction of the boring device. As shown in FIG. **11**, the cooling liquid feed mechanism **160** includes: a rotor **161** including the shank **82** gripped by the first gripping mechanism **18** of the electric drill **11**; and a cooling liquid feeder **163** disposed around the rotor **161**. The cooling liquid feeder **163** can be kept from rotating by a bearing **164** disposed between the cooling liquid feeder **163** and the rotor **161**. The feed flow path **88** is connected to the cooling liquid feeder **163**. A feed flow path **163a** disposed inside the cooling liquid feeder **163** communicates with a feed flow path **161a** disposed inside the rotor **161**.

(64) At the front of the rotor **161** is disposed a bit retainer **166** that grips the proximal end portion of the drill bit **90**. The proximal end portion of the drill bit **90** is inserted into and held by the bit retainer **166**. The mechanism for holding the proximal end portion of the drill bit **90** is omitted in the drawings. Any known technical means can be used as the mechanism for holding the proximal end portion of the drill bit **90**.

(65) Inside the rotor **161** is disposed a sliding member **167** movable a predetermined distance in the front-rear direction (i.e., the longitudinal direction of the drill bit **90**) while maintaining contact with the proximal end portion of the drill bit **90**. The sliding member **167** is biased forward by a biasing member **168** (spring) disposed inside the rotor **161**. At the rear of the sliding member **167** is disposed an incline **167a** along which the diameter of the sliding member **167** increases. The biasing force of the biasing member **168** keeps the incline **167a** in contact with a sealing member **165** disposed at a given location on the rotor **161**. The front portion of the sliding member **167** is sealed by a sealing member **169** disposed between the front portion of the sliding member **167** and the bit retainer **166**. O-rings can be used as the sealing members **165** and **169**. The sliding member **167** is slidable rearward against the biasing force of the biasing member **168** within a given size of gap **A** provided between the sliding member **167** and the rotor **161**.

(66) A space **161b** between the sliding member **167** and the rotor **161** communicates with a cooling liquid feed hole **92** of the drill bit **90** via a feed flow path **167b** of the sliding member **167**. The cooling liquid feed hole **92** extends from the proximal end portion of the drill bit **90** to the cutting edge **98**.

(67) As shown in FIG. 11, in the state where the sliding member **167** is biased forward by the biasing member **168**, the incline **167a** is in contact with the sealing member **165**, and the feed flow path **161a** and the space **161b** are disconnected from each other. In this state, the cooling liquid L fed from the feed flow path **88** to the cooling liquid feeder **163** does not flow from the feed flow path **161a** to the space **161b**.

(68) Once the sliding member **167** is pushed from the direction of the drill bit **90** (the state shown in FIG. 14B described later), the sliding member **167** moves rearward against the biasing force of the biasing member **168** within the gap A of given size. In this state, the incline **167a** is away from the sealing member **165**, and the feed flow path **161a** and the space **161b** are connected, so that the cooling liquid L fed from the feed flow path **88** flows from the feed flow path **161a** to the space **161b**.

(69) With the drill bit **90** attached to the bit retainer **166**, the cooling liquid L is fed to and stopped in the cooling liquid feed mechanism **160**. Once the drill bit **90** is pushed rearward, the cooling liquid L in the cooling liquid feed mechanism **160** is delivered to the cutting edge **98** through the cooling liquid feed hole **92** of the drill bit **90** and fed from the cutting edge **98** to the boring target area Wa.

(70) In other words, the boring device **10D** according to the present embodiment further includes: a sliding member **167** mounted on the proximal end of the drill bit **90** and slidable together with the drill bit **90** in the longitudinal direction of the drill bit **90**; and a biasing member **168** that biases the drill bit **90** and the sliding member **167** in the direction from the proximal end of the drill bit **90** toward the cutting edge **98**. The sliding member **167** closes the feed flow path **88** when placed toward the cutting edge **98** by the biasing force of the biasing member **168**, and opens the feed flow path **88** when slid proximally against the biasing force of the biasing member **168**.

(71) In this embodiment, the gap A of given size, within which the sliding member **167** moves in the front-rear direction, is smaller than a gap B (see FIGS. 11 and 14B) provided between the front surface of the shield **200** biased forward by the biasing mechanism **170** described later and the cutting edge **98** of the drill bit **90**. Thus, the incline **167a** of the sliding member **167** moves away from the sealing member **165** after the front surface of the shield **200** is first brought into contact with the workpiece W and then the drill bit **90** is pushed in the gap B by a distance corresponding to the size of the gap A. This allows a sealing member **208** of the shield **200** to be pressed against the workpiece W while the cooling liquid L is fed from the cutting edge **98** (see FIG. 14B), thus preventing leakage of the cooling liquid L from a space between the shield **200** and the workpiece W. The above-described relationship between the gaps A and B is merely an example, and the relationship between the gaps A and B is not limited to that in this embodiment.

(72) Shield **200**

(73) FIGS. 12A and 12B are each a schematic view of the shield of the boring device according to the present embodiment. FIG. 12A is a schematic view of the shield as viewed from the proximal end of the drill bit, and FIG. 12B is a cross-sectional view of the shield cut at the center in the width direction of the boring device along the longitudinal direction of the boring device. The shield **200** of this embodiment includes a horizontally-long main body **201** and couplers **202** disposed at the two longitudinal ends of the main body **201**. Sliders **175** of the biasing mechanism **170** shown in FIG. 13 are coupled to the couplers **202**. At the center of the main body **201** is disposed a guide member **203** that guides the front portion of the drill bit **90** in the front-rear direction. A guide **204** for the drill bit **90** is disposed at the center of the guide member **203**, and a cavity **205** is disposed in the front portion of the guide member **203**. A sealing member **206** is disposed between the guide member **203** and the main body **201** to prevent leakage of the suspension S from a space between the guide member **203** and the main body **201**. An O-ring can be used as the sealing member **206**.

(74) At the front surface of the main body **201** is disposed the sealing member **208** that makes contact with the workpiece W. The sealing member **208** seals the periphery of the cavity **205** and

further seals the periphery of the boring target area Wa. A sponge or rubber material can be used as the sealing member **208**.

(75) At the lower portion of the main body **201** is located a discharge hole **207** leading from the cavity **205** to the exterior of the main body **201**. The discharge flow path **138** is connected to the discharge hole **207**. The suspension S in the cavity **205** is discharged into the discharge flow path **138** through the discharge hole **207**.

(76) Biasing Mechanism **170**

(77) FIG. **13** is a schematic view showing a biasing mechanism of the boring device according to the fourth embodiment of the present invention. FIG. **13** is a half-sectional view. The biasing mechanism **170** of this embodiment includes a main body **171** mounted on the cooling liquid feed mechanism **160** of the boring device **10D**. The main body **171** is secured to the outer surface of the cooling liquid feed mechanism **160**.

(78) The main body **171** is provided with two guides **173** extending in the front-rear direction and two sliders **175** slidable along the guides **173** in the front-rear direction, the guides **173** and sliders **175** being located to the left and right of the cooling liquid feed mechanism **160**. The guides **173** are inserted into holes of the main body **171** and fastened at opposite left and right points by fastening bolts **172**, thus being secured to the main body **171**. The sliders **175** are biased forward by biasing springs **174** disposed inside the guides **173**. The forward sliding of the sliders **175** is prevented by holders **176** disposed at the front end portions of the guides **173**, and the sliders **175** are held at given positions as shown in FIG. **13**. The sliders **175** are slidable rearward from the shown positions against the biasing forces of the biasing springs **174**.

(79) The shield **200** is secured to the front ends of the sliders **175**. Thus, the shield **200** is movable rearward together with the sliders **175** against the biasing forces of the biasing springs **174**.

(80) Example of Boring Operation

(81) FIGS. **14A** and **14B** are each a cross-sectional view of the cooling liquid feed mechanism and its neighboring components of the boring device according to the present embodiment and show the cooling liquid feed mechanism and its neighboring components cut at the center in the width direction of the boring device along the longitudinal direction of the boring device. FIG. **14A** is a cross-sectional view of the cooling liquid feed mechanism and its neighboring components in a state with the feed flow path closed before a boring operation, and FIG. **14B** is a cross-sectional view of the cooling liquid feed mechanism and its neighboring components in a state with the feed flow path open during the boring operation. An exemplary boring operation performed on the workpiece W using the boring device **10D** and separation device **140** will be described based on FIGS. **10** and **14**.

(82) First, as shown in FIG. **10**, the boring device **10D** is connected to the separation device **140** via the feed flow path **88** and the discharge flow path **138**. The pump unit **20'** and drill bit **90** are driven by the drive unit **13**, and the cutting edge **98** of the boring device **10D** is placed on the boring target area Wa of the workpiece W. In this state, the tube pump **60'** presses the feed tube **62** in an attempt to feed the cooling liquid L from the liquid vessel **141** to the cutting edge **98** of the drill bit **90**.

However, as shown in FIG. **14A**, the sliding member **167** of the cooling liquid feed mechanism **160** is in contact with the sealing member **165** and closes the end of the feed flow path **161a**, and thus the cooling liquid L is not fed from the liquid vessel **141**. The tube pump **60'** presses the discharge tube **112**, and this leads to a state where air has been sucked into the discharge flow path **138** from the cavity **205** (FIG. **12B**) of the shield **200**. In this state, the sealing member **208** of the shield **200** of the boring device **10D** is placed against the boring target area Wa of the workpiece W.

(83) Subsequently, as shown in FIG. **14B**, an operation interface **12a** disposed on the housing **12** is operated to rotate the drill bit **90**, and the cutting edge **98** of the rotating drill bit **90** is pressed against the workpiece W. This leads to the shield **200** being pressed against the workpiece W with a suitable force by the biasing mechanism **170**. As a result of the cutting edge **98** of the drill bit **90** being pressed against the workpiece W, the proximal end portion of the drill bit **90** causes the

sliding member **167** to move rearward against the biasing force of the biasing member **168** within the gap A of given size.

(84) Thus, the incline **167a** of the sliding member **167** moves away from the sealing member **165**, and the feed flow path **163a** of the cooling liquid feeder **163** is connected to the cooling liquid feed hole **92** of the drill bit **90** through the feed flow path **161a** and space **161b** of the rotor **161** and the feed flow path **167b** of the sliding member **167**. As such, the cooling liquid L delivered from the liquid vessel **141** to the cooling liquid feed mechanism **160** is fed from the cutting edge **98** of the drill bit **90** to the boring target area Wa while a hole is being bored in the workpiece W by the cutting edge **98** of the drill bit **90**. The suspension S in the cavity **205** (see FIG. 12B) of the shield **200** is forcibly discharged into the liquid vessel **141** as a result of the discharge tube **112** being pressed by the pressing mechanism **64** (**114**) (see FIG. 10).

(85) As described above, the cutting edge **98** of the rotating drill bit **90** is pressed against the workpiece W located ahead of the cutting edge **98**, and a hole is bored in the boring target area Wa of the workpiece W by the cutting edge **98**. During this boring operation, the feed tube **62** is pressed by the pressing mechanism **64** (**114**), and accordingly the cooling liquid L in the liquid vessel **141** is forcibly delivered to the cutting edge **98** of the boring device **10D** and fed from the cutting edge **98** to the boring target area Wa. At the same time, the discharge tube **112** is pressed by the pressing mechanism **64** (**114**), and accordingly the suspension S is forcibly discharged out of the shield **200** into the liquid vessel **141**.

(86) Once the suspension S is discharged into the liquid vessel **141**, the chips C and other substances settle under their own weight. Additionally, the strainer **147** floating in the vicinity of the liquid surface removes the chips C. Thus, the cooling liquid L is recovered and fed to the cutting edge **98** again. That is, when the cooling liquid L to be fed to the cutting edge **98** of the drill bit **90** attached to the boring device **10D** is recovered from the suspension S held in the liquid vessel **141**, the subsurface portion of the suspension S, which contains a smaller amount of chips C than the rest of the suspension S, is passed through the strainer **147**. As such, the cooling liquid L little contaminated by the chips C is circulated to the cutting edge **98** of the boring device **10D**. Furthermore, the use of the tube pump **60'** allows for proper circulation of the cooling liquid L even if the cooling liquid L is somewhat contaminated by the chips C.

(87) Thus, with the use of the boring device **10D** described above, cooling liquid circulation with reliable backflow prevention can be accomplished by combination of proper feed of the cooling liquid L to the cutting edge **98** and forcible discharge of the suspension S, and at the same time a proper boring operation can be performed on the workpiece W.

(88) Additionally, since the boring device **10D** described above forcibly discharges the suspension S out of the shield **200** covering the boring target area Wa, the cooling liquid L can be circulated properly even in a boring operation during which the cutting edge is oriented downward. Boring operations with varying orientations of the cutting edge can be performed properly.

(89) In the present embodiment, a cart for carrying the separation device **140** may be employed. This allows for easy transfer of the entire system. The transfer of the entire system can be easily accomplished using a backpack or shoulder bag instead of the cart. In the case where the electric drill **11** is equipped with an internal battery **19** as shown in FIG. 10, the system can be easily transferred with the separation device **140** placed on the cart. This makes it easier to change the site where a boring operation is to be performed. Additionally, the boring operation can be properly performed even if the workplace is relatively narrow. The electric drill **11** of the boring device **10D** is not limited to being equipped with the internal battery **19**. The electric drill **11** may be supplied with electric power through a power cord **19a**, like the previously-described boring devices **10A** to **10C** (see FIGS. 1, 8, and 9).

(90) Variants

(91) From the foregoing description, numerous modifications and other embodiments of the present invention are obvious to those skilled in the art. Accordingly, the foregoing description is to be

construed as illustrative only, and is provided for the purpose of teaching those skilled in the art the best mode for carrying out the present invention. The structural and/or functional details may be substantially modified without departing from the gist of the present invention.

(92) In the first to fourth embodiments described above, the speed reduction mechanism **30** includes the two planetary gear mechanisms **31a** and **31b** aligned in the axial direction of the input shaft **21**. This is not limiting and, for example, the speed reduction mechanism **30** may include three or more planetary gear mechanisms aligned in the axial direction of the input shaft **21**.

(93) In this case, each of the three or more planetary gear mechanisms may, like the planetary gear mechanisms **31a** and **31b**, include a sun gear, three planetary gears, and an internal gear. In this case, the planetary carrier is disposed between each pair of adjacent planetary gear mechanisms of the three or more planetary gear mechanisms. The planetary carrier is connected to the plurality of planetary gears of one of the adjacent planetary gear mechanisms that is closer to the drive unit, and rotates together with the sun gear of the other of the adjacent planetary gear mechanisms that is closer to the cutting edge.

(94) In the first to fourth embodiments described above, the sun gear **34a** projects from, and is integrated with, the outer surface of the input shaft **21**. This is not limiting and, for example, the sun gear of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms need not be integrated with the outer surface of the input shaft. In this case, for example, the sun gear may include a shaft hole, and the inner wall of the shaft hole may be secured to the outer surface of the input shaft to secure the sun gear to the outer surface of the input shaft.

(95) In the first to fourth embodiments described above, the planetary gear mechanism **31a** includes the three planetary gears **36a**, and the planetary gear mechanism **31b** includes the three planetary gears **36b**. This is not limiting and, for example, each of the plurality of planetary gear mechanisms may include two planetary gears or four or more planetary gears. Each of the plurality of planetary gear mechanisms may include a different number of planetary gears.

(96) In the first to fourth embodiments described above, the sun gears **34a** and **34b** are coaxial with the input shaft **21**. This is not limiting and, for example, appropriate addition of another gear can eliminate the need for the sun gears to be coaxial with the input shaft.

(97) In the first to fourth embodiments described above, the planetary carrier **50** includes the three fitting holes **52** (fitting-receiving portions) into which the respective rotational shafts **37a** of the three planetary gears **36a** are fitted. This is not limiting and, for example, the planetary carrier may include recesses serving as the fitting-receiving portions into which the respective rotational shafts of the plurality of planetary gears are fitted.

(98) In the first to fourth embodiments described above, the speed reduction mechanism **30** includes the two planetary gear mechanisms **31a** and **31b** aligned in the axial direction of the input shaft **21**. This is not limiting and, for example, the speed reduction mechanism may be constructed as Harmonic Drive (registered trademark).

(99) In the above case, the first pressing mechanism may include: a rotational plate (or rotor) mounted on the input shaft of Harmonic Drive (registered trademark) to be rotatable in a plane perpendicular to the input shaft; and at least one presser disposed on the periphery of the rotational plate (or rotor). The first pressing mechanism may further include a cover including an inner wall internally contacted by the at least one presser, with the feed tube (and discharge tube) interposed between the inner wall and the at least one presser, to press the feed tube (and discharge tube) in cooperation with the at least one presser. The tube pump thus constructed can also force the cooling liquid in the feed flow path to flow to the cutting edge (and force the suspension S in the discharge flow path to flow downstream of the discharge flow path).

(100) In the fourth embodiment described above, a single pressing mechanism **64** (**114**) serves as both the first pressing mechanism **64** that presses the feed tube **62** and the second pressing mechanism **114** that presses the discharge tube **112**, and the feed tube **62** and discharge tube **112** are pressed by the same three rollers **66** (see FIGS. 5 and 6). This is not limiting and, for example, the

first and second pressing mechanisms may be constructed separately, and the discharge tube may be pressed by pressers disposed separately from pressers that press the feed tube.

Summary

(101) In order to solve the problem previously described, a boring device according to one embodiment of the present invention includes: a cutting edge; a drive unit that rotationally drives the cutting edge; a feed flow path through which a cooling liquid is fed to the cutting edge; and a tube pump that forces the cooling liquid in the feed flow path to flow to the cutting edge, wherein the tube pump includes a feed tube constituting a part of the feed flow path and a first pressing mechanism that presses the feed tube, and the first pressing mechanism is driven by the drive unit.

(102) With the above features, the tube pump is driven by the drive unit that rotationally drives the cutting edge. Thus, the cooling liquid can be forced to flow to the cutting edge with a simple device configuration.

(103) The boring device may further include a speed reduction mechanism disposed between the drive unit and the first pressing mechanism to reduce the speed of rotation produced by power derived from the drive unit and transmit the resulting power to the first pressing mechanism.

(104) With the above features, the first pressing mechanism can be rotated at a lower speed than the cutting edge. This makes it possible to accomplish a boring operation successfully.

(105) For example, the boring device may further include an input shaft through which the power derived from the drive unit is input to the speed reduction mechanism. The input shaft may extend from the drive unit and penetrate at least the speed reduction mechanism toward the cutting edge, and the speed reduction mechanism may include a plurality of planetary gear mechanisms aligned in an axial direction of the input shaft and a planetary carrier connecting a pair of adjacent planetary gear mechanisms of the plurality of planetary gear mechanisms. Each of the plurality of planetary gear mechanisms may include a sun gear, a plurality of planetary gears externally contacting the sun gear to rotate in mesh with the sun gear, and an internal gear internally contacted by the plurality of planetary gears to rotate in mesh with the plurality of planetary gears. The sun gear of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms may rotate together with the input shaft, and the planetary carrier may be connected to the plurality of planetary gears of one of the adjacent planetary gear mechanisms that is closer to the drive unit, and rotate together with the sun gear of the other of the adjacent planetary gear mechanisms that is closer to the cutting edge. The first pressing mechanism may include: a plurality of pressers mounted respectively on the plurality of planetary gears of the planetary gear mechanism closest to the cutting edge among the plurality of planetary gear mechanisms, the pressers being rotatable together with the plurality of planetary gears, respectively; and a cover including an inner wall internally contacted by the plurality of pressers, with the feed tube interposed between the inner wall and the plurality of pressers, to press the feed tube in cooperation with the plurality of pressers.

(106) The sun gears of the plurality of planetary gear mechanisms may be coaxial with the input shaft, the sun gear of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms may be secured to the input shaft, and each of the sun gears of the planetary gear mechanisms other than the planetary gear mechanism closest to the drive unit may include a first shaft hole through which the input shaft extends.

(107) With the above features, the configuration of the boring device can be further simplified.

(108) The planetary carrier may include: a plurality of fitting-receiving portions into which are fitted respective rotational shafts of the plurality of planetary gears of one of the adjacent planetary gear mechanisms that is closer to the drive unit; and a second shaft hole through which the input shaft extends.

(109) For example, the boring device may further include a housing enclosing at least the drive unit.

(110) The tube pump may be disposed outside the housing.

(111) With the above feature, the tube pump can be easily attached to and detached from the drive unit.

(112) The tube pump may be enclosed in the housing.

(113) With the above feature, the occurrence of failure of the tube pump can be reduced since the tube pump is not exposed to the outside environment.

(114) The cutting edge may be a distal end of a drill bit, and the boring device may further include: the drill bit; a sliding member mounted on a proximal end of the drill bit and slidable together with the drill bit in a longitudinal direction of the drill bit; and a biasing member that biases the drill bit and the sliding member in a direction from the proximal end of the drill bit toward the cutting edge. The sliding member may close the feed flow path when placed toward the cutting edge by a biasing force of the biasing member and open the feed flow path when slid proximally against the biasing force of the biasing member.

(115) With the above features, when the cutting edge is pressed against a workpiece in a boring operation, the sliding member slides proximally against the biasing force of the biasing member to open the feed flow path. Thus, the boring operation can be performed efficiently.

(116) The boring device may further include a discharge flow path through which a suspension is discharged from the cutting edge, and the tube pump may further include: a discharge tube constituting a part of the discharge flow path; and a second pressing mechanism that presses the discharge tube. The second pressing mechanism may be driven by the drive unit to force the suspension in the discharge flow path to flow downstream of the discharge flow path.

(117) With the above features, the suspension can be forcibly discharged downstream of the discharge flow path from the cutting edge. Thus, the suspension can be reliably discharged downstream of the discharge flow path even in the case where the boring target area of the workpiece is located at a lower height than the downstream end of the discharge flow path or where the boring operation is performed with the cutting edge oriented downward.

(118) A single pressing mechanism may serve as both the first and second pressing mechanisms, and the single pressing mechanism may press the feed tube and the discharge tube to force the cooling liquid in the feed flow path to flow to the cutting edge and force the suspension in the discharge flow path to flow downstream of the discharge flow path.

(119) With the above features, feed of the cooling liquid to the cutting edge and discharge of the suspension from the cutting edge can be efficiently achieved with a simple construction.

(120) An upstream end of the feed flow path and a downstream end of the discharge flow path may be connected to a separation device that separates chips of a workpiece from the suspension to recover the cooling liquid.

(121) With the above feature, the cooling liquid can be circulated. Thus, boring operations can be performed repeatedly without having to prepare a new cooling liquid for each operation.

REFERENCE CHARACTERS LIST

(122) **10A to 10D** boring device **11** electric drill **12** housing **13** drive unit **14** electric motor **15** drive shaft **16** motor speed reduction mechanism **18** first gripping mechanism **20** pump unit **21** input shaft **22, 82** shank **30** speed reduction mechanism **31a, 31b** planetary gear mechanism **32** first circular plate **33, 58, 69, 73** through hole **34a, 34b** sun gear **35, 54** shaft hole **36a, 36b** planetary gear **37a, 37b** rotational shaft **38a, 38b** internal gear **50** planetary carrier **52, 68** fitting hole **56** second circular plate **60, 60'** tube pump **62** feed tube **64** first pressing mechanism **66** roller (presser) **67** third circular plate **70** cover **72** inner wall **75, 75'** insertion hole **76** outlet hole **80** second gripping mechanism **88** feed flow path **90** drill bit **92** cooling liquid feed hole **98** cutting edge **112** discharge tube **114** second pressing mechanism **138** discharge flow path **140** separation device **160** cooling liquid feed mechanism **167** sliding member **168** biasing member **L** cooling liquid **S** suspension **W** workpiece

Claims

1. A boring device comprising: a cutting edge; a drive unit that rotationally drives the cutting edge; a feed flow path through which a cooling liquid is fed to the cutting edge; and a tube pump that forces the cooling liquid in the feed flow path to flow to the cutting edge, wherein the tube pump includes a feed tube constituting a part of the feed flow path and a first pressing mechanism that presses the feed tube, and the first pressing mechanism is driven by the drive unit.
2. The boring device according to claim 1, further comprising a speed reduction mechanism disposed between the drive unit and the first pressing mechanism to reduce the speed of rotation produced by power derived from the drive unit and transmit the resulting power to the first pressing mechanism.
3. The boring device according to claim 2, further comprising an input shaft through which the power derived from the drive unit is input to the speed reduction mechanism, wherein the input shaft extends from the drive unit and penetrates at least the speed reduction mechanism toward the cutting edge, the speed reduction mechanism includes a plurality of planetary gear mechanisms aligned in an axial direction of the input shaft and a planetary carrier connecting a pair of adjacent planetary gear mechanisms of the plurality of planetary gear mechanisms, each of the plurality of planetary gear mechanisms includes a sun gear, a plurality of planetary gears externally contacting the sun gear to rotate in mesh with the sun gear, and an internal gear internally contacted by the plurality of planetary gears to rotate in mesh with the plurality of planetary gears, the sun gear of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms rotates together with the input shaft, the planetary carrier is connected to the plurality of planetary gears of one of the adjacent planetary gear mechanisms that is closer to the drive unit, and rotates together with the sun gear of the other of the adjacent planetary gear mechanisms that is closer to the cutting edge, and the first pressing mechanism includes: a plurality of pressers mounted respectively on the plurality of planetary gears of the planetary gear mechanism closest to the cutting edge among the plurality of planetary gear mechanisms, the pressers being rotatable together with the plurality of planetary gears, respectively; and a cover including an inner wall internally contacted by the plurality of pressers, with the feed tube interposed between the inner wall and the plurality of pressers, to press the feed tube in cooperation with the plurality of pressers.
4. The boring device according to claim 3, wherein the sun gears of the plurality of planetary gear mechanisms are coaxial with the input shaft, the sun gear of the planetary gear mechanism closest to the drive unit among the plurality of planetary gear mechanisms is secured to the input shaft, and each of the sun gears of the planetary gear mechanisms other than the planetary gear mechanism closest to the drive unit includes a first shaft hole through which the input shaft extends.
5. The boring device according to claim 3, wherein the planetary carrier includes: a plurality of fitting-receiving portions into which are fitted respective rotational shafts of the plurality of planetary gears of one of the adjacent planetary gear mechanisms that is closer to the drive unit; and a second shaft hole through which the input shaft extends.
6. The boring device according to claim 1, further comprising a housing enclosing at least the drive unit.
7. The boring device according to claim 6, wherein the tube pump is disposed outside the housing.
8. The boring device according to claim 6, wherein the tube pump is enclosed in the housing.
9. The boring device according to claim 1, wherein the cutting edge is a distal end of a drill bit, the boring device further comprises: the drill bit; a sliding member mounted on a proximal end of the drill bit and slidable together with the drill bit in a longitudinal direction of the drill bit; and a biasing member that biases the drill bit and the sliding member in a direction from the proximal end of the drill bit toward the cutting edge, and the sliding member closes the feed flow path when

placed toward the cutting edge by a biasing force of the biasing member and opens the feed flow path when slid proximally against the biasing force of the biasing member.

10. The boring device according to claim 1, further comprising a discharge flow path through which a suspension is discharged from the cutting edge, wherein the tube pump further includes: a discharge tube constituting a part of the discharge flow path; and a second pressing mechanism that presses the discharge tube, and the second pressing mechanism is driven by the drive unit to force the suspension in the discharge flow path to flow downstream of the discharge flow path.

11. The boring device according to claim 10, wherein a single pressing mechanism serves as both the first and second pressing mechanisms, and the single pressing mechanism presses the feed tube and the discharge tube to force the cooling liquid in the feed flow path to flow to the cutting edge and force the suspension in the discharge flow path to flow downstream of the discharge flow path.

12. The boring device according to claim 11, wherein an upstream end of the feed flow path and a downstream end of the discharge flow path are connected to a separation device that separates chips of a workpiece from the suspension to recover the cooling liquid.
